

Planar Graphs

Lecture Notes – Advanced Algorithms 2026

These notes follow the lecture slides *Planar Graphs: Planarity Testing, Planar Separator Theorem + Applications*. They cover the same material with the arguments written out in full, together with the informal reasoning that usually gets spoken rather than projected. Sections marked (*non-examinable*) go beyond what was presented.

Contents

1	Introduction	2
2	Basics	2
2.1	Planar graphs, plane graphs, faces	2
2.2	Euler’s formula	3
2.3	Consequences: edge counts and low-degree vertices	4
2.4	Minors and the Kuratowski–Wagner theorem	5
2.5	Colouring planar graphs	7
2.6	Drawing with straight lines	8
3	Planarity testing	8
3.1	Warm-up: showing $K_{3,3}$ is not planar by hand	9
3.2	Reducing to the 2-connected case	12
3.3	The key insight	12
3.4	Segments and interlacement	13
3.5	The algorithm	14
3.6	The catch, and separating cycles	14
3.7	Running time	16
4	The Planar Separator Theorem	17
4.1	The bound is tight	18
4.2	Koebe’s theorem and the proof idea (<i>non-examinable</i>)	18
5	Application: independent set in $2^{\mathcal{O}(\sqrt{n})}$ time	19
6	What stays hard on planar graphs	22
6.1	Planar Clique is easy	22
6.2	Planar 3-Colouring is NP-hard	22
7	Summary	24
8	Exercises	24

1 Introduction

A graph is planar if it can be drawn in the plane without edges crossing. That is a purely topological condition, and at first sight it has nothing to do with algorithms. It turns out to be one of the most algorithmically consequential structural restrictions we know.

There are two separate reasons to care, and this lecture is organised around them.

The first is that planarity is a real constraint that real inputs satisfy. Road networks, printed circuit boards, floor plans, and many geometric meshes are planar or nearly so. If your input is planar you would like to know it, and you would like to exploit it. So the first algorithmic question is simply: *given a graph, can we decide efficiently whether it is planar?* The answer is yes, and remarkably it can be done in linear time; we will study a slower but far more instructive $\mathcal{O}(n^3)$ algorithm, because the ideas in it — cycles, segments, interlacement — are the ideas everybody uses.

The second reason is deeper. Planar graphs cannot be tangled: they have small *balanced separators*. Deleting only $\mathcal{O}(\sqrt{n})$ vertices splits a planar graph into pieces none of which is larger than a constant fraction of the whole. This is the Planar Separator Theorem, and it is the engine behind a whole family of divide-and-conquer algorithms. We will use it to solve MAXIMUM INDEPENDENT SET — NP-hard in general, and still NP-hard on planar graphs — in time $2^{\mathcal{O}(\sqrt{n})}$ rather than $2^{\mathcal{O}(n)}$. That is not polynomial, but on a graph with 10 000 vertices it is the difference between impossible and routine.

We close by looking at what *stays* hard. Most NP-hard graph problems remain NP-hard when restricted to planar inputs, and we prove this for 3-colouring. There are a few notable exceptions, and CLIQUE is one of them — for a reason that is a one-line consequence of the structure theory in Section 2.

- Section 2: Euler’s formula and its consequences, minors, Kuratowski–Wagner, colouring, Fáry’s theorem.
- Section 3: an $\mathcal{O}(n^3)$ planarity test.
- Section 4: the Planar Separator Theorem.
- Section 5: independent set in $2^{\mathcal{O}(\sqrt{n})}$ time.
- Section 6: what remains NP-hard, and what does not.

Throughout, $G = (V, E)$ is a graph with $n = |V|$ vertices and $m = |E|$ edges.

2 Basics

2.1 Planar graphs, plane graphs, faces

We must distinguish two notions that are easy to conflate, and the distinction matters constantly in what follows.

Definition 2.1 (Planar and plane). A graph is *planar* if it *can* be drawn in $\mathbb{R}^2$ with no two edges crossing. A *plane graph* is a planar graph *together with* one such drawing: the embedding is fixed.

Planarity is a property of an abstract graph; being plane is a property of a drawing. The same planar graph has many essentially different embeddings, and several of the quantities we are about to define — faces, in particular — depend on the embedding rather than on the graph.

Definition 2.2 (Faces). Let G be a plane graph. Removing the drawing from the plane leaves a collection of connected open regions, called the *faces* of the embedding. Exactly one of them is unbounded; it is the *outer face*, and the others are the *interior faces*.

[figure placeholder]

Left: a graph drawn with crossings. Right: the same graph redrawn without crossings, with the faces shaded and labelled $f_1, \dots, f_k$, and the outer face marked. The point of the picture is that the left drawing does not prove the graph is non-planar — it only proves that this drawing is bad.

Figure 1: Planar is a property of the graph; the faces belong to the drawing.

A useful bookkeeping fact, with one exception that will bite us if we ignore it: *every edge is incident to exactly two faces* — except a *bridge* (an edge whose removal disconnects its component), which has the same face on both sides. The right way to think about it, and the way that makes the counting arguments below uniform, is to count *edge-sides*: every edge has two sides, and each side lies on exactly one face. For a non-bridge these are two different faces; for a bridge they are the same face twice.

[figure placeholder]

A small plane graph containing a bridge. Highlight the bridge and shade the single face that touches it on both sides, next to a normal edge with its two distinct faces shaded in different colours.

Figure 2: An edge has two sides; a bridge has the same face on both of them.

2.2 Euler's formula

Theorem 2.3 (Euler). *Let G be a connected plane graph with $n \geq 1$ vertices, m edges and f faces. Then*

$$n + f = m + 2.$$

Proof. By induction on m .

Base case $m = 0$. A connected graph with no edges is a single vertex, so $n = 1$, and the drawing leaves the whole plane as one face, $f = 1$. Indeed $1 + 1 = 0 + 2$.

Inductive step. Let $m \geq 1$ and assume the formula for all connected plane graphs with fewer edges. We distinguish two cases, and we must check that they are exhaustive.

- **G contains a cycle.** Let e be an edge on that cycle. Then e is not a bridge, so the two sides of e belong to two *different* faces. Deleting e keeps the graph connected and merges those two faces into one. So the new graph has $n' = n$, $m' = m - 1$ and $f' = f - 1$, and it is still a connected plane graph. By induction $n' + f' = m' + 2$, that is $n + f - 1 = m - 1 + 2$, which rearranges to $n + f = m + 2$.

- **G has a vertex v of degree 1.** Delete v and its edge. The graph stays connected, no faces merge or split (the pendant edge had the same face on both sides), so $n' = n - 1$, $m' = m - 1$, $f' = f$. By induction $n - 1 + f = m - 1 + 2$, i.e. $n + f = m + 2$.

These two cases cover everything: if G has no vertex of degree 1 and at least one edge, then — being connected — it has minimum degree at least 2, and a graph with minimum degree at least 2 contains a cycle. (Walk forward without immediately reusing the edge you came on; since the graph is finite you must eventually revisit a vertex, and the walk between the two visits is a cycle.) $\square$

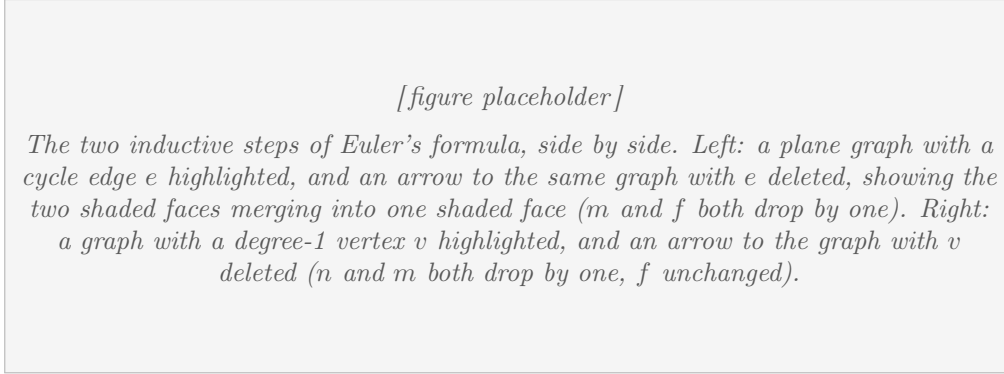


Figure 3: Deleting a cycle edge merges two faces; deleting a pendant vertex changes no face.

Intuition. Euler's formula says that a plane drawing has no slack: the three quantities n , m , f are locked together. This is what turns a topological hypothesis ("drawable without crossings") into a combinatorial one ("not too many edges"), and every quantitative statement in this section is an application of that translation.

Notice also that n and m do not depend on the drawing, so neither does f : *every* plane embedding of a given connected planar graph has the same number of faces. That is already a non-trivial corollary.

2.3 Consequences: edge counts and low-degree vertices

Euler's formula alone does not bound m ; we also need that faces cannot be too small. That is where simplicity comes in.

Corollary 2.4. *Let G be a connected plane graph with no parallel edges and no self-loops and with $n > 2$. Then*

$$m \leq 3n - 6.$$

Proof. Count edge-sides in two ways. Every face is bounded by at least three edge-sides: with no loops and no parallel edges and at least three vertices, no face can be enclosed by one or two edge-sides. Summing over faces gives a total of at least $3f$ edge-sides. On the other hand every edge has exactly two sides, so the total is exactly $2m$. Hence

$$3f \leq 2m, \quad \text{i.e.} \quad f \leq \frac{2}{3}m.$$

Substituting into Euler's formula $f = m + 2 - n$ gives

$$m + 2 - n \leq \frac{2}{3}m \quad \implies \quad \frac{1}{3}m \leq n - 2 \quad \implies \quad m \leq 3n - 6. \quad \square$$

Corollary 2.5 (K_5 is not planar). *The complete graph K_5 is not planar.*

Proof. K_5 has $n = 5$ and $m = 10$, but $3n - 6 = 9 < 10$. □

This is the first genuinely useful thing we can do: a purely arithmetic certificate of non-planarity. It does *not* work for $K_{3,3}$, which has $n = 6$, $m = 9 \leq 12$; there one has to use that bipartite plane graphs have faces of length at least four, giving $m \leq 2n - 4 = 8 < 9$. Try it as an exercise. A completely different and much more instructive proof, which is really the planarity-testing algorithm run by hand, is given in Section 3.1.

Corollary 2.6 (Sparse graphs have sparse spots). *Every plane graph with no parallel edges and no self-loops has a vertex of degree at most 5.*

Proof. For $n \leq 2$ this is trivial, so assume $n > 2$ and, by taking a component, that G is connected. The average degree is

$$\frac{1}{n} \sum_{v \in V} \deg(v) = \frac{2m}{n} \leq \frac{2(3n - 6)}{n} = 6 - \frac{12}{n} < 6.$$

A finite set of integers with average less than 6 must contain an element that is at most 5. □

Intuition. Corollary 2.6 is small but it is the workhorse of the whole subject. It says planar graphs always have a cheap place to start an induction. Every colouring result below is “delete a low-degree vertex, colour the rest, put the vertex back”, and the only reason that strategy is available at all is this corollary. Note too that it is inherited: every subgraph of a planar graph is planar, so the induction never leaves the class.

2.4 Minors and the Kuratowski–Wagner theorem

Corollary 2.5 suggests a plan: characterise planarity by forbidden sub-structures. The right notion of sub-structure is not the subgraph but the minor.

Definition 2.7 (Minor). A graph H is a *minor* of a graph G if H can be obtained from G by a sequence of the following operations:

- *edge deletion*,
- *vertex deletion* (together with its incident edges),
- *edge contraction*: replace an edge $\{u, v\}$ by a single new vertex adjacent to all former neighbours of u and of v .

The single most important property, and the reason minors are the right notion here, is this:

Observation 2.8. *Every minor of a planar graph is planar.*

Proof idea. Each of the three operations can be performed directly on a drawing. Deleting an edge or a vertex only erases ink. Contracting an edge $\{u, v\}$ can be done by sliding u along the drawn edge towards v , dragging its other edges with it; since we slide along a curve that is already crossing-free, no crossings are created. □

So the class of planar graphs is *minor-closed*, and any minor of a planar graph is a legitimate certificate: exhibiting a non-planar minor proves non-planarity. Two small examples of what minors can express:

[figure placeholder]

IMPORTANT. Reproduce the edge-contraction example from the slides. Left: a graph on vertices $a, b, c, d, e, f, g, h, i$ with the edge $\{d, e\}$ highlighted. Right: the result of contracting it, with d and e merged into a single vertex labelled de inheriting all their edges. Keep the same layout on both sides so the eye can follow what moved.

Figure 4: Edge contraction: $\{d, e\}$ is contracted to a single vertex de .

Example 2.9. K_3 (a triangle) is a minor of G if and only if G contains a cycle. Indeed, contracting a cycle of length $\ell > 3$ once shortens it to length $\ell - 1$, so any cycle can be contracted down to a triangle; conversely a triangle minor certainly requires a cycle to have been present.

Theorem 2.10 (Kuratowski, Wagner). *A graph is planar if and only if it has neither K_5 nor $K_{3,3}$ as a minor.*

We do not prove this. One direction is easy and worth stating explicitly: if G is planar then by Observation 2.8 all its minors are planar, and K_5 (Corollary 2.5) and $K_{3,3}$ (the exercise above) are not. The hard direction is that these two obstructions are the *only* ones, which is a genuinely deep statement — it is the prototype for Robertson and Seymour’s graph minor theorem, which says every minor-closed class has a finite set of forbidden minors.

[figure placeholder]

K_5 and $K_{3,3}$ drawn side by side, each in its standard picture, labelled “the only two obstructions”. Perhaps also show the $K_{3,3}$ edge count $m = 9 > 8 = 2n - 4$ as an annotation.

Figure 5: The two forbidden minors.

Remark 2.11 (Why this does not give an algorithm). Theorem 2.10 is a characterisation, not a procedure. To use it directly we would have to search for a K_5 or $K_{3,3}$ minor, and the number of candidate minors is exponential — we would have to choose which vertices get merged into which branch set. This is exactly why Section 3 builds a completely different algorithm. (It is *true* that testing for a fixed minor H can be done in polynomial time by deep results of Robertson and Seymour, but that machinery is far heavier than the problem deserves, and it came thirty years later.)

[figure placeholder]

IMPORTANT — classroom exercise. Reproduce the two “planar or not?” graphs from the slides. These are meant to be worked on live: students should either find a crossing-free drawing or exhibit a $K_5 / K_{3,3}$ minor. Leave enough white space next to each for students to redraw.

Figure 6: Planar or not? Decide for each, and justify with a drawing or a forbidden minor.

2.5 Colouring planar graphs

Definition 2.12. A k -colouring of G is a map $c : V \rightarrow \{1, \dots, k\}$ such that $c(u) \neq c(v)$ for every edge $\{u, v\} \in E$. G is k -colourable if such a map exists.

Proposition 2.13. Every planar graph is 6-colourable.

Proof. Induction on $|V|$. Graphs with at most 6 vertices are trivially 6-colourable. Otherwise, by Corollary 2.6 there is a vertex v with $\deg(v) \leq 5$. The graph $G - v$ is planar and smaller, so by induction it has a 6-colouring. Now put v back: its at most 5 neighbours use at most 5 of the 6 colours, so at least one colour is free for v . $\square$

The same idea gives one colour better, but the last step needs an extra idea: with $\deg(v) = 5$ all six colours may be blocked, so we must arrange for two of v 's neighbours to share a colour *before* we remove v .

Theorem 2.14. Every planar graph is 5-colourable.

Proof. Induction on $|V|$; graphs with at most 5 vertices are trivial. Let G be planar with $n > 5$ and let v be a vertex with $\deg(v) \leq 5$ (Corollary 2.6).

Case 1: $\deg(v) \leq 4$. Exactly as in Proposition 2.13: 5-colour $G - v$ by induction, and give v one of the at most 4 colours not used by its neighbours.

Case 2: $\deg(v) = 5$. We first claim that v has two non-adjacent neighbours. Suppose not: then all five neighbours of v are pairwise adjacent, so they form a K_5 subgraph of G , contradicting Corollary 2.5. Fix two non-adjacent neighbours x and y .

Let G' be obtained from G by deleting v and *identifying* x with y . Then G' is a minor of G : contract the two edges $\{v, x\}$ and $\{v, y\}$ — which merges x, v and y into a single vertex — and delete the edges from v to its three other neighbours. By Observation 2.8, G' is planar. It also has $n - 2 < n$ vertices, so by induction it has a 5-colouring c' .

Pull c' back to $G - v$: every vertex other than x, y keeps its colour, and both x and y receive the colour of the merged vertex. This is a proper colouring of $G - v$, because the only new adjacency the identification could have created is between x and y themselves, and x, y are non-adjacent in G by choice.

Finally colour v . Its five neighbours include x and y , which now have the same colour, so they use at most 4 distinct colours between them. One of the five colours is free, and we give it to v . $\square$

[figure placeholder]

IMPORTANT. The five-colour argument in three panels. Panel 1: vertex v of degree exactly 5 with its five neighbours drawn around it, two of them x and y marked as non-adjacent (dashed “no edge” between them). Panel 2: G' , with v gone and x, y merged into one vertex, drawn so that planarity is visibly preserved. Panel 3: the colouring pulled back — x and y carry the same colour, the five neighbours of v show only four distinct colours, and v takes the leftover colour.

Figure 7: Identifying two non-adjacent neighbours frees up a colour for v .

Intuition. The whole difficulty of 5-colouring is concentrated in one place: a degree-5 vertex can see five different colours, and then induction stalls. The fix is to force a collision in advance. We are allowed to force it because planarity guarantees that two of the neighbours are non-adjacent, so gluing them together is harmless; and we are allowed to glue at all because gluing is a minor operation and minors of planar graphs stay planar. Both licences come from Section 2, which is why that section came first.

Theorem 2.15 (Four Colour Theorem; Appel–Haken 1976). *Every planar graph is 4-colourable.*

We will not prove this, and neither will anybody else on a blackboard. The proof reduces the statement to a finite but enormous case analysis — an unavoidable set of configurations, each checked by computer. It was the first major theorem whose proof could not be verified by a human unaided, and it caused a lasting argument about what a proof is for. (A formally verified proof, machine-checked end to end, was completed by Gonthier in 2005, which settles the correctness question if not the philosophical one.)

Note that 4 is optimal: K_4 is planar and needs four colours. And in the other direction, deciding 3-colourability of a planar graph is NP-hard — we prove that in Section 6.2. So the complexity landscape for planar colouring is completely mapped: $k \leq 2$ easy, $k = 3$ NP-hard, $k \geq 4$ always yes.

2.6 Drawing with straight lines

Theorem 2.16 (Fáry, 1936). *Every planar graph can be drawn in the plane with no crossings and with every edge drawn as a straight line segment.*

This is reassuring rather than surprising: it says the curved edges we allow ourselves are a convenience, not a necessity. It is also the entry point to a large research area, *graph drawing*, which asks for drawings of (almost) planar graphs that are good in various additional senses — small area on an integer grid, few distinct edge slopes, all faces convex, vertices at prescribed positions, few crossings when the graph is not quite planar.

3 Planarity testing

Problem. Given a graph G , decide whether it is planar (and if so, produce an embedding).

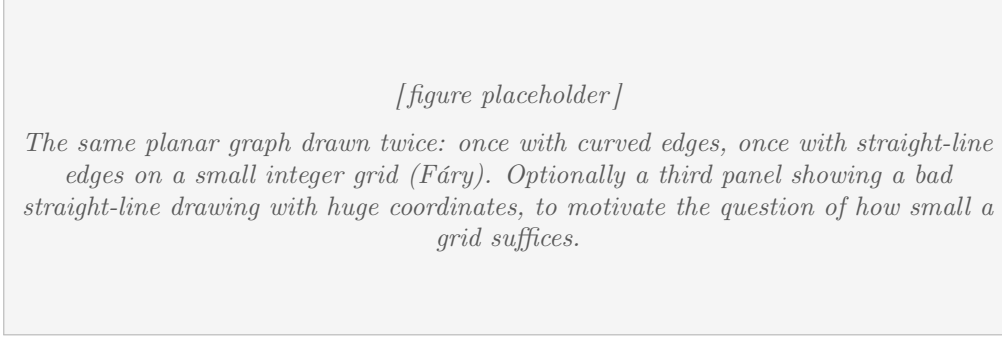


Figure 8: Fáry's theorem: curves are never necessary.

Linear-time algorithms are known — the classical ones are Hopcroft–Tarjan (1974) and Booth–Lueker (1976), and there are several later ones that are easier to implement. They are also fiddly, and their correctness proofs are long. We will instead study an $\mathcal{O}(n^3)$ algorithm going back to Auslander–Parter and Goldstein, because it is built from ideas you can hold in your head, and because those ideas (cycle, segments, interlacement) reappear in the fast algorithms anyway.

As noted in Remark 2.11, Kuratowski's theorem is of no direct help. We need a constructive strategy.

3.1 Warm-up: showing $K_{3,3}$ is not planar by hand

Before any formalism, let us do the entire thing on the smallest interesting example. Everything the general algorithm does will appear here in miniature, and it is worth returning to this subsection whenever the machinery in Sections 3.4–3.6 feels abstract.

We already know from the edge count $m \leq 2n - 4$ that $K_{3,3}$ is not planar. That proof is shorter than the one below, but it is a dead end: it tells us nothing about how to *search* for a drawing. The argument here is longer and much more useful, because it is a procedure.

Write the two sides of $K_{3,3}$ as $\{a_1, a_2, a_3\}$ and $\{b_1, b_2, b_3\}$.

Step 1: draw a cycle. Start with any cycle and commit to drawing it. Take the 6-cycle

$$C = a_1 b_1 a_2 b_2 a_3 b_3 a_1,$$

which uses six of the nine edges of $K_{3,3}$ (check each consecutive pair: it always joins an a to a b , so it is indeed a cycle in the graph). Drawn without crossings, C is a closed curve, and it cuts the plane into an inside and an outside.

Step 2: see what is left over. Three edges remain:

$$a_1 b_2, \quad a_2 b_3, \quad a_3 b_1.$$

Each of them joins two vertices that are already on C and touches C nowhere else. So each must be drawn either entirely inside C or entirely outside C — there is no third option, and no edge can be split between the two regions. We have turned a drawing problem into three yes/no decisions.

Step 3: see which decisions clash. Number the positions around C as

$$\underbrace{a_1}_1 \underbrace{b_1}_2 \underbrace{a_2}_3 \underbrace{b_2}_4 \underbrace{a_3}_5 \underbrace{b_3}_6,$$

so the three leftover edges join positions

$$a_1b_2 : 1-4, \quad a_2b_3 : 3-6, \quad a_3b_1 : 5-2.$$

Any two of these three *interleave*: going around the cycle, their endpoints alternate. For 1–4 and 3–6, the order encountered is 1, 3, 4, 6 — one endpoint of the second chord lies between the endpoints of the first, and the other lies outside. The same is true for 1–4 against 5–2, and for 3–6 against 5–2.

Two interleaving chords cannot be drawn on the same side of C : inside the disc bounded by C , two curves whose endpoints alternate around the boundary are forced to cross. For the moment we take this as visually evident — it is exactly Definition 3.3 and Observation 3.5 below, where it is stated properly.

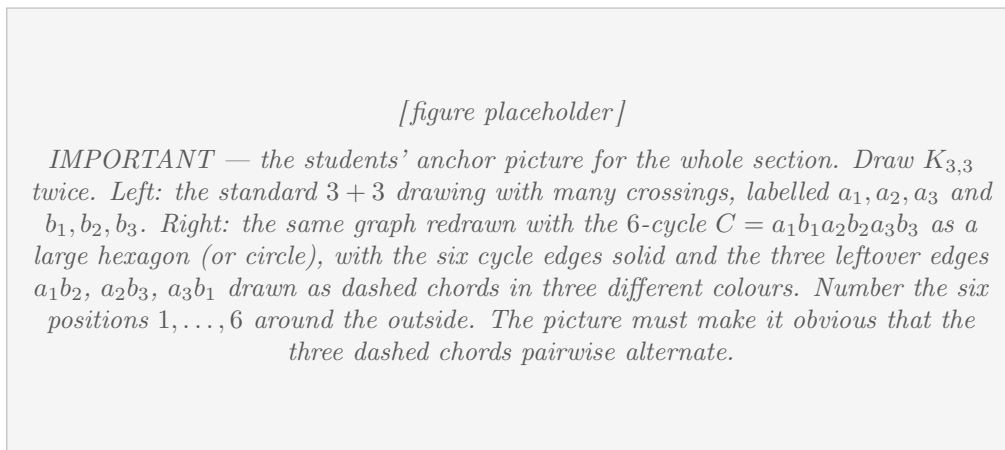


Figure 9: $K_{3,3}$ redrawn around a 6-cycle. The three remaining edges are chords, and each must go inside or outside.

Step 4: get stuck. Now try to make the three decisions.

- Put a_1b_2 *inside*. (Why we may assume this is the subtle point; see below.)
- a_2b_3 interleaves with a_1b_2 , so it cannot also be inside. It must go *outside*.
- a_3b_1 interleaves with a_1b_2 , so it cannot go inside. It also interleaves with a_2b_3 , so it cannot go outside. There is nowhere left to put it.

We are stuck, and therefore no crossing-free drawing exists: $K_{3,3}$ is not planar. □

Why the first choice is free. The one step that needs justification is putting a_1b_2 inside. We did not prove that it belongs there; we assumed it. This is legitimate because the two cases are the *same* case.

Suppose we had a crossing-free drawing in which a_1b_2 lies outside C . Project the drawing stereographically onto the sphere. On the sphere the curve C still separates the surface into two regions, but neither of them is distinguished — there is no “outer” region,

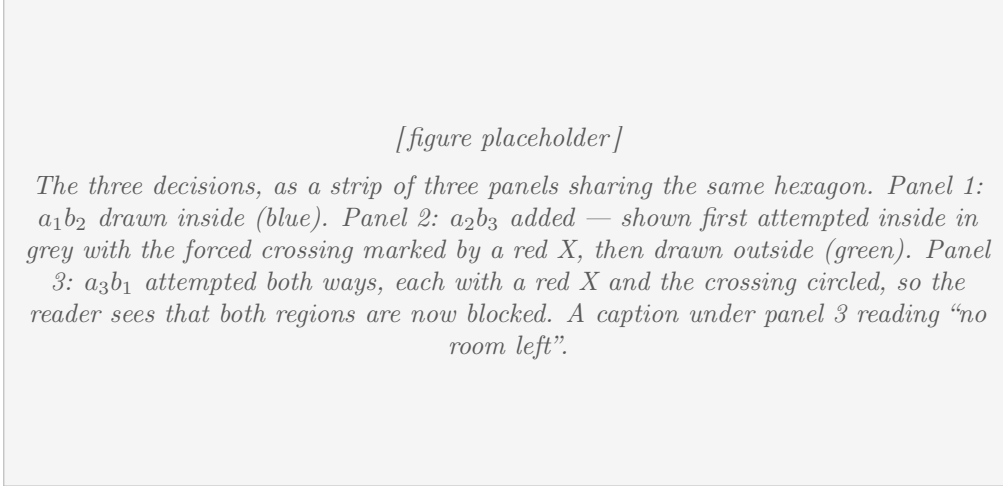


Figure 10: Placing the three chords: the third one has nowhere to go.

only two discs. Rotate the sphere so that the two regions trade places, and project back to the plane. The result is again a crossing-free drawing of the same graph with the same cycle C , but now a_1b_2 is inside. (Equivalently, and without leaving the plane: invert the drawing through a circle, which exchanges the interior and exterior of C .)

So any drawing can be converted into one where a_1b_2 is inside, and it is enough to rule out that case. After that first free choice nothing else is free: each subsequent chord’s side is forced.

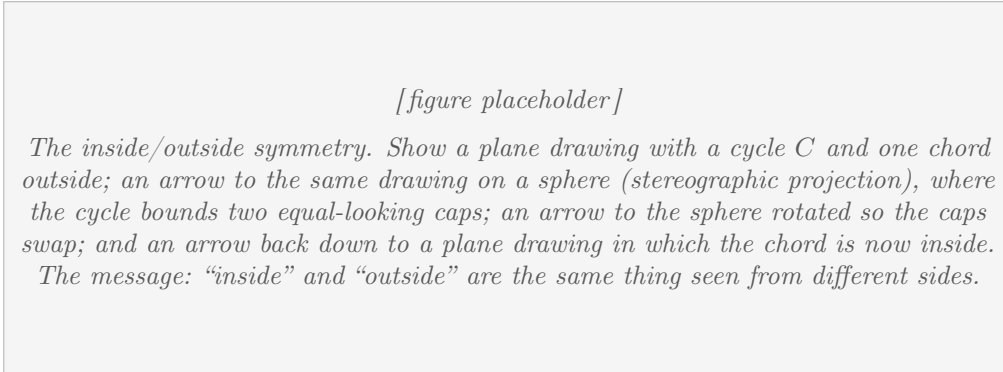


Figure 11: On the sphere there is no outer face, so the first chord may be placed inside without loss of generality.

Intuition. That was the whole algorithm. Read the four steps again and note what each one becomes in general:

In the $K_{3,3}$ warm-up	In the general algorithm
the 6-cycle C	a <i>separating cycle</i> C (Section 3.6)
the three leftover chords	the <i>segments</i> of C (Definition 3.2)
“these two chords interleave”	the <i>interlacement graph</i> (Definition 3.3)
inside / outside for each chord	a 2-colouring of that graph
“we got stuck”	the interlacement graph is not 2-colourable

There is exactly one genuinely new ingredient in the general case. A chord is a single edge, so once we decide which side it goes on there is nothing more to check. A segment

is a whole subgraph, and deciding to put it inside C leaves the question of whether it can be drawn there — which is another planarity problem. That is where the recursion comes from, and it is the only thing the warm-up does not already show you.

3.2 Reducing to the 2-connected case

Two easy preprocessing steps let us assume the input is well-behaved.

- **Connectivity.** G is planar if and only if each of its connected components is planar: separate components can be drawn far apart in the plane and never interfere. So test each component.
- **2-connectivity.** If v is a cut vertex, split G at v into the pieces formed by the components of $G - v$ together with v itself. G is planar if and only if all the pieces are. Informally, one can rotate each piece around v so that it occupies its own wedge of the plane touching v ; nothing forces two pieces to interleave, because they meet in the single point v .

Iterating the second step decomposes G into its *blocks* (maximal 2-connected subgraphs), computable in linear time by depth-first search, and G is planar exactly when all of its blocks are.

So from now on: **G is 2-connected.** Concretely we use two consequences: every vertex lies on a cycle, and between any two vertices there are two internally disjoint paths.

Finally, one free sanity check that will matter for the running time: by Corollary 2.4, if $m > 3n - 6$ we may immediately answer NO. So we may assume $m = \mathcal{O}(n)$ throughout.

3.3 The key insight

Everything rests on one observation about what a cycle does to a plane drawing.

Observation 3.1 (The key insight). *Let C be a cycle in a plane graph G . A cycle drawn without crossings is a closed curve, and by the Jordan curve theorem it divides the plane into an inside and an outside. Consequently, if $\{u, v\} \in E$ is an edge with neither u nor v on C , then u and v lie both inside C or both outside C — an edge cannot get from one region to the other without crossing C .*

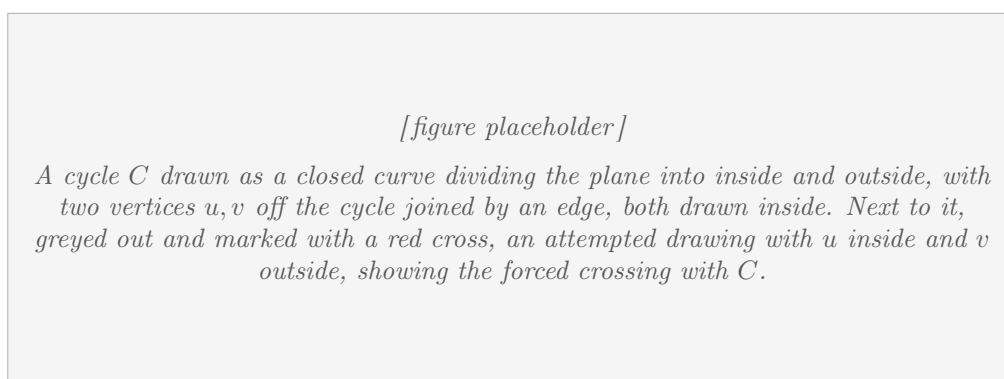


Figure 12: A cycle separates the plane: edges avoiding it cannot cross it.

So once we commit to drawing a cycle C , the rest of the graph decomposes into chunks, and each chunk must be placed entirely inside or entirely outside. This turns a continuous drawing problem into a discrete two-way choice — and two-way choices are exactly what 2-colouring solves.

3.4 Segments and interlacement

Definition 3.2 (Segment). Fix a cycle C in G . A *segment* (also called a *bridge* of C) is either

- an edge of G joining two non-consecutive vertices of C — a *chord*; or
- the subgraph induced by the edges of one connected component of $G - V(C)$, together with all edges joining that component to C .

The vertices of C that a segment touches are its *attachments*.

By Observation 3.1, each segment must be drawn entirely inside C or entirely outside C ; it cannot be split, because it is connected and avoids C except at its attachments.

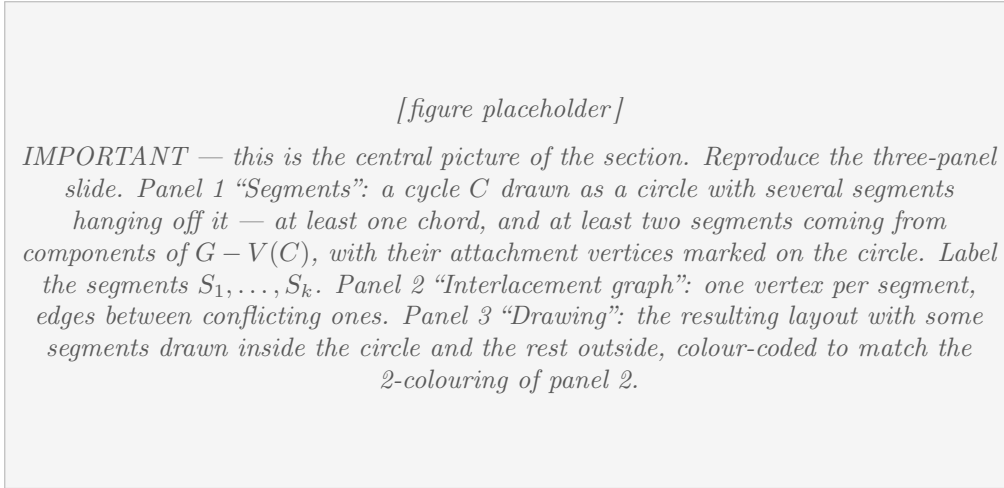


Figure 13: From a cycle to segments to the interlacement graph to a drawing.

Now, when can two segments be placed on the same side? If their attachments interleave around the cycle, they cannot: drawing both inside would force them to cross.

Definition 3.3 (Conflict / interlacement). Two segments S and S' *conflict* (or *interlace*) if there are four vertices v_1, v_2, v_3, v_4 occurring in this cyclic order on C with v_1, v_3 attachments of S and v_2, v_4 attachments of S' . The *interlacement graph* I has one vertex per segment, with an edge between two segments exactly when they conflict.

Remark 3.4 (A case the slide glosses over). There is a second way two segments can fail to share a side: if they have *three or more attachments in common*. Suppose S and S' both attach at a, b, c . Drawing S inside C cuts the disc into three regions, each of whose boundary contains only two of a, b, c ; so S' , which must live in one region, cannot reach all three. These pairs are not always caught by Definition 3.3 (the four witnesses would have to repeat a vertex), so a correct implementation must add them to I explicitly. This is standard in the literature; see the optional Auslander–Parter reading in the course archive.

Observation 3.5. *If G is planar, then in any plane embedding the segments placed inside C form an independent set of I , as do those placed outside. Hence I must be 2-colourable, i.e. bipartite.*

This gives us a necessary condition that we can test in linear time (2-colourability is a BFS). It is not by itself sufficient — a segment placed inside must also be drawable inside, which is a planarity question one level down — and that is what makes the algorithm recursive.

[figure placeholder]

Two panels. Left: two segments whose attachments alternate around the cycle (v_1, v_2, v_3, v_4 in cyclic order, S attaching at v_1, v_3 and S' at v_2, v_4), with an attempt to draw both inside showing the forced crossing in red. Right: two segments whose attachments do not alternate, drawn happily side by side inside the cycle. A third small panel could show the three-common-attachments case of Remark 3.4.

Figure 14: Interlacing segments cannot share a side.

3.5 The algorithm

Algorithm 1 Planarity test (first attempt)

```

1: find a cycle  $C$  in  $G$ ; let  $S_1, \dots, S_k$  be its segments
2: if the interlacement graph  $I$  is not 2-colourable then
3:   return NO
4: end if
5: for  $i = 1, \dots, k$  do
6:   recursively test whether  $C + S_i$  is planar
7: end for
8: return YES iff all recursive calls returned YES

```

Why is this the right recursion? Two directions.

- **Soundness.** If I is not 2-colourable then G is not planar by Observation 3.5. If some $C + S_i$ is not planar, then G is not planar either, since $C + S_i$ is a subgraph of G .
- **Completeness.** Suppose I is 2-colourable and each $C + S_i$ is planar. Take a 2-colouring of I ; it tells us which segments go inside and which go outside. Each $C + S_i$ has a plane embedding, which we may take with S_i drawn inside C (if it is drawn outside, invert the drawing through a circle, or think of it on the sphere and rotate). Now merge: draw C once, and for each i paste in the drawing of S_i , on the side dictated by the colouring. Two segments on the same side do not conflict, so their attachments do not interleave and their drawings can be arranged in disjoint parts of the disc without crossings.

3.6 The catch, and separating cycles

Algorithm 1 has a bug, and it is a fatal one: the recursion need not make progress. If C happens to have only one segment S_1 , then $C + S_1 = G$, and we recurse on the very same graph forever.

The fix is to insist on a cycle that actually splits something off.

Definition 3.6. A cycle C in G is *separating* if it has at least two segments.

[figure placeholder]

The merging step. Show the cycle C drawn once in the middle, and around it several separately-computed drawings of $C + S_i$, each with S_i inside its own copy of C . Arrows show them being pasted into the common picture, blue ones inside the circle and red ones outside, matching the 2-colouring. Convey that the drawings only ever meet on C itself.

Figure 15: Merging the recursive drawings using the 2-colouring.

If C is separating, then every $C + S_i$ is a proper subgraph of G (it misses at least the edges of the other segment), so the recursion strictly descends. It remains to show that separating cycles exist, except in cases so simple that we can answer directly.

Lemma 3.7 (Existence of a separating cycle). *Let G be 2-connected. Then G has a separating cycle unless*

- (i) G is itself a cycle, or
- (ii) G is a cycle plus one segment which is a path.

Moreover, a separating cycle can be found in polynomial (indeed linear) time.

Both exceptional cases are trivially planar: (i) is a circle, and (ii) is a *theta graph*, drawn as a circle with one path across the inside. So the lemma loses us nothing.

Proof. Find any cycle C (one exists since G is 2-connected). If C has at least two segments we are done. If C has no segments then $G = C$ and we are in case (i). So assume C has exactly one segment S . If S is a path, we are in case (ii). So assume S is a single segment that is *not* a path; we construct a separating cycle.

Since G is 2-connected, S has at least two attachments on C . Pick two attachments u and v that are *consecutive*, meaning that one of the two u – v arcs of C contains no attachment of S in its interior. Call that arc B , and call the other arc A ; so A contains all the remaining attachments of S , if any.

Because S is connected and attaches at both u and v , it contains a path P from u to v whose interior vertices lie entirely inside S (in particular, off C). Set

$$C' := A + P,$$

which is a cycle: A is a u – v path along C , P is a u – v path through S , and they meet only at u and v .

We claim C' is separating, i.e. that it has at least two segments.

- The arc B gives one segment of C' : its interior vertices are off C' and connect to C' exactly at u and v . (If B has no interior vertices then B is a single edge uv , which is a chord of C' because P has length at least 2; either way, a segment.)
- Since S is not a path, S has at least one edge outside P . That edge is not on C' either, so it belongs to some segment of C' , and that segment is different from the one contributed by B — it lives on the other side, attached to $A \cup P$ rather than only to $\{u, v\}$.

So C' has at least two segments and is separating. All the steps — finding a cycle, finding attachments, finding P — are searches in a graph and run in linear time. $\square$

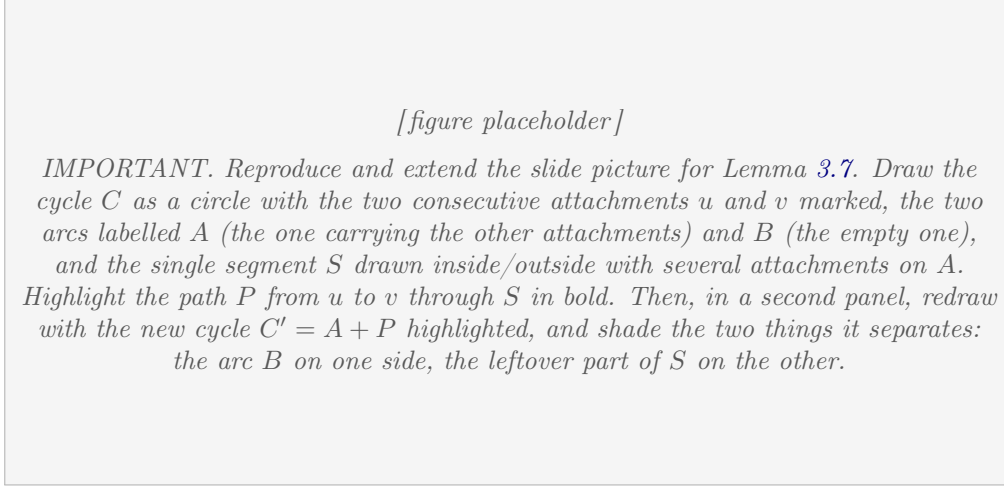


Figure 16: Building a separating cycle $C' = A + P$ when C has a single non-path segment.

Algorithm 2 Planarity test (corrected)

```

1: reduce to the 2-connected case; if  $m > 3n - 6$  return NO
2: if  $G$  is a cycle, or a cycle plus a path segment then
3:   return YES
4: end if
5: find a separating cycle  $C$  (Lemma 3.7); let  $S_1, \dots, S_k$  be its segments
6: if the interlacement graph  $I$  is not 2-colourable then
7:   return NO
8: end if
9: for  $i = 1, \dots, k$  do
10:   recursively test whether  $C + S_i$  is planar
11: end for
12: return YES iff all recursive calls returned YES

```

3.7 Running time

Each recursive call does a bounded amount of graph work: find a separating cycle (linear), compute the segments (a connected-components computation, linear), build the interlacement graph (at most $\mathcal{O}(n^2)$, since there are $\mathcal{O}(n)$ segments and comparing two segments' attachment patterns takes linear time), and 2-colour it (linear in its size, $\mathcal{O}(n^2)$). So a call costs $\mathcal{O}(n^2)$.

How many calls are there? Each call passes a strictly smaller graph $C + S_i$ to its children, but the pieces overlap in C , so the naive bound is not immediate.

Fact (stated without proof). The total number of recursive calls is at most m . The argument charges each call to a distinct edge of G ; the details are omitted here as in the lecture.

Since we may assume $m = \mathcal{O}(n)$, we get $\mathcal{O}(n)$ calls of cost $\mathcal{O}(n^2)$ each:

$$\text{total running time } \mathcal{O}(n^3).$$

Intuition. Step back and look at the shape of this algorithm. It converts a *geometric* question — can this be drawn? — into a *combinatorial* one — can these choices be made consistently? The conversion happens in two moves. Observation 3.1 says the only choice is binary (inside or outside). Definition 3.3 records which binary choices are incompatible. What is left is 2-colouring, which is easy, plus a recursion to check that each individual choice can actually be realised.

The one place where care is genuinely needed is the recursion measure, which is also the one place where the naive version was simply wrong. That is a common pattern: the interesting bug is never in the clever idea, it is in the termination argument.

4 The Planar Separator Theorem

We now turn to the structural property that makes planar graphs algorithmically soft: they can be cut into balanced pieces very cheaply.

Definition 4.1 (Balanced separator). Let $\alpha \in (0, 1)$. A set $S \subseteq V$ is an α -balanced separator of G if every connected component of $G - S$ has at most $\alpha \cdot n$ vertices.

Note what is *not* required: $G - S$ may have many components, and they need not be balanced against each other. All we ask is that no single component is large.

Theorem 4.2 (Planar Separator Theorem). Let $G = (V, E)$ be a planar graph on n vertices. Then G has a $\frac{9}{10}$ -balanced separator of size at most $11\sqrt{n}$. Moreover, such a separator can be found in polynomial time.

Several proofs exist. The classical one is due to Lipton and Tarjan (1979), who obtained a $\frac{2}{3}$ -balanced separator of size $2\sqrt{2}\sqrt{n}$ using breadth-first search layers; it is combinatorial and constructive. The proof sketched below is geometric and probabilistic, and it generalises better. The exact constants are of no importance for us — what matters is $\mathcal{O}(\sqrt{n})$.

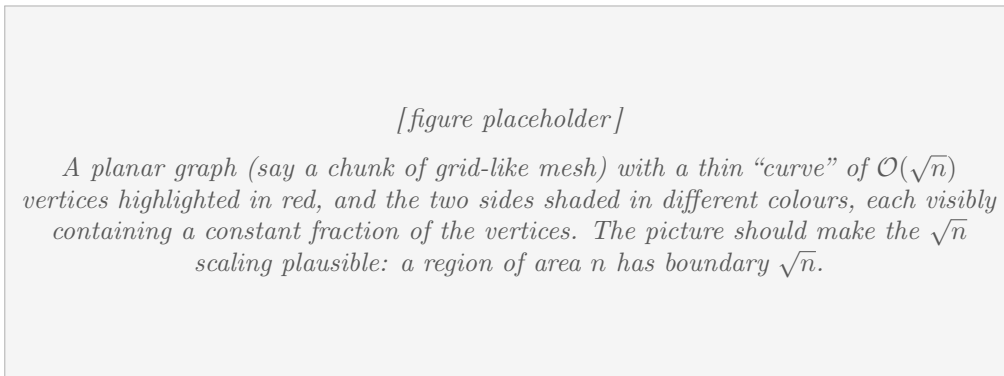


Figure 17: A balanced separator: remove $\mathcal{O}(\sqrt{n})$ vertices, and nothing big survives.

4.1 The bound is tight

Proposition 4.3. *The $\sqrt{n} \times \sqrt{n}$ grid has no $\frac{9}{10}$ -balanced separator of size $\sqrt{n}/20$.*

So Theorem 4.2 is optimal up to the constant factor, and no cleverer argument will get us to, say, $\mathcal{O}(n^{1/3})$.

Intuition. The grid is the canonical hard case and it is worth carrying around as a mental image. To break a $\sqrt{n} \times \sqrt{n}$ grid into pieces of at most $\frac{9}{10}$ of the area you must cut it more or less all the way across, and going all the way across costs about $\sqrt{n}$ vertices. Any set that is too small leaves a “highway” of untouched rows and columns, and the untouched part stays connected and large.

The same picture explains why $\sqrt{n}$ is the right exponent in general: planar graphs are two-dimensional, and in two dimensions the boundary of a region of area A has length about $\sqrt{A}$. The Planar Separator Theorem is the combinatorial shadow of the isoperimetric inequality.

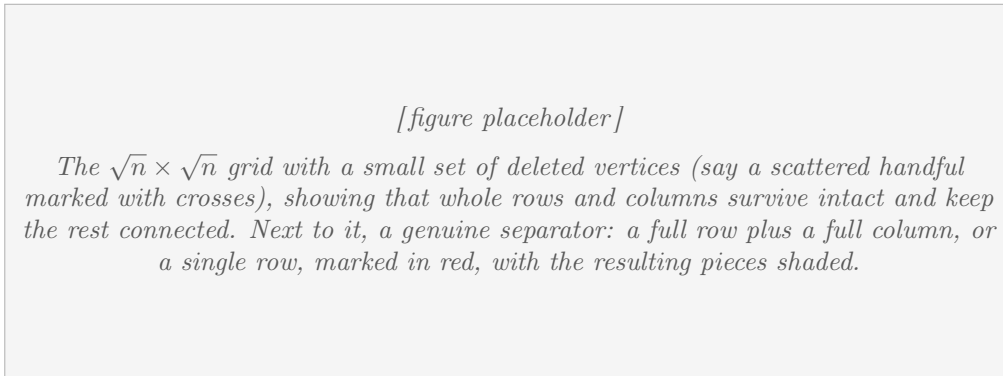


Figure 18: Small sets cannot disconnect the grid.

4.2 Koebe’s theorem and the proof idea (non-examinable)

The geometric proof rests on a beautiful and surprising representation theorem.

Theorem 4.4 (Koebe’s disk embedding theorem). *Every planar graph G can be represented by a family of disks $\{D_v\}_{v \in V}$ in the plane such that*

- (i) *the interiors of the disks are pairwise disjoint, and*
- (ii) *D_v and D_w touch if and only if $\{v, w\} \in E$.*

In other words, every planar graph *is* a coin-contact graph. This converts a combinatorial object into a rigid geometric one, at which point geometric tools — areas, random circles — become available.

Sketch, for Theorem 4.2. This was omitted in the lecture and is optional reading. Fix a disk representation of G as given by Theorem 4.4.

Let D be the smallest circle containing at least $n/10$ of the disk centres. Rescale and translate so that D has radius 1 and is centred at the origin. Pick $r \in [1, 2]$ uniformly at random, let C_r be the circle of radius r about the origin, and set

$$S = \{v : D_v \text{ intersects } C_r\},$$

a random set. Take S as the separator; the vertices whose disks lie strictly inside C_r and those strictly outside form the two sides.

[figure placeholder]

A small planar graph next to its circle packing: disks of various sizes, touching exactly where the graph has edges, with the graph drawn faintly on top (vertices at disk centres, edges through the tangency points). Include a few large disks and many small ones so the size variation is visible — that variation is what the proof exploits.

Figure 19: Koebe: every planar graph is the contact graph of a family of disks.

Why S separates. Two disks that do not meet C_r and lie on opposite sides of it cannot touch, so there is no edge between the inside part and the outside part. This is the same Jordan-curve reasoning as Observation 3.1, now applied to a circle drawn through the packing.

Why S is balanced. The inside contains at least the $n/10$ centres inside D . For the outside, one shows that the disk of radius 2 cannot contain too many centres: by minimality of D , every circle of radius smaller than 1 contains fewer than $n/10$ centres, and a disk of radius 2 can be covered by a constant number of such circles. Choosing the constants appropriately gives that neither side exceeds $\frac{9}{10}n$.

Why S is small. This is the pretty part. A disk D_v of radius ρ_v meets C_r only if r falls in an interval of length about $2\rho_v$, so

$$\Pr[v \in S] \lesssim 2\rho_v,$$

because r is uniform on an interval of length 1. Hence $\mathbb{E}[|S|] \lesssim 2 \sum_v \rho_v$, where the sum ranges over the disks that could possibly be hit — essentially those inside the disk of radius 2 (only $\mathcal{O}(1)$ disks are large enough to reach it from outside). Those disks are interior-disjoint and fit inside a bounded region, so $\sum_v \pi \rho_v^2 = \mathcal{O}(1)$, and by Cauchy–Schwarz

$$\sum_v \rho_v \leq \sqrt{n \cdot \sum_v \rho_v^2} = \mathcal{O}(\sqrt{n}).$$

So $\mathbb{E}[|S|] = \mathcal{O}(\sqrt{n})$, and some choice of r achieves the expectation. Working out the constants carefully gives the $11\sqrt{n}$ of Theorem 4.2. $\square$

Intuition. The two requirements on S pull in opposite directions, and the randomness is what reconciles them. A circle far out is likely to cut few disks but leaves an unbalanced split; a circle close in balances well but may slice through a lot. Averaging over a whole range of radii shows that a good compromise must exist, without ever having to point at it. This is the probabilistic method in its purest form: we prove a separator exists by computing an average, and we never construct one by hand.

5 Application: independent set in $2^{\mathcal{O}(\sqrt{n})}$ time

MAXIMUM INDEPENDENT SET is NP-hard, and it stays NP-hard on planar graphs. So we should not hope for a polynomial algorithm. But the separator theorem buys us an enormous speed-up over the trivial 2^n .

[figure placeholder]

The probabilistic argument. A circle packing with the small circle D (radius 1, containing $n/10$ centres) marked in the middle, the annulus between radius 1 and radius 2 shaded, and one sampled circle C_r drawn in red. Highlight the disks that C_r cuts through — these form S . Convey visually that C_r can only cut a disk if the disk is big or if r lands in a narrow window, which is why big disks are rare and small disks are rarely hit.

Figure 20: A random circle in the annulus cuts few disks and splits the rest evenly.

Theorem 5.1. *MAXIMUM INDEPENDENT SET on planar graphs can be solved in $2^{\mathcal{O}(\sqrt{n})}$ time.*

Intuition. The idea is guess-and-divide. If we knew which separator vertices the optimal solution uses, the problem would fall apart into independent subproblems, one per component of $G - S$ — because there are no edges between different components, so their choices cannot interfere. We do not know, so we try all possibilities. There are only $2^{|S|} = 2^{\mathcal{O}(\sqrt{n})}$ of them, which is affordable, and each guess leaves subproblems that are a constant factor smaller. The recursion depth is $\mathcal{O}(\log n)$ and the sizes shrink geometrically, so the $\sqrt{n}$ in the exponent survives to the end.

Algorithm 3 MAX-INDEPENDENT-SET(G) for planar G

```

1: if  $n$  is below a fixed constant then return a maximum independent set by brute force
2: end if
3: find a  $\frac{9}{10}$ -balanced separator  $S$  with  $|S| \leq 11\sqrt{n}$  ▷ Theorem 4.2
4: let  $G_1, \dots, G_t$  be the connected components of  $G - S$ 
5:  $X \leftarrow \emptyset$ 
6: for all independent sets  $I$  of  $G[S]$  do
7:   for  $i = 1, \dots, t$  do
8:      $I_i \leftarrow \text{MAX-INDEPENDENT-SET}(G_i - N(I))$ 
9:   end for
10:   $Y \leftarrow I \cup I_1 \cup \dots \cup I_t$ 
11:  if  $|Y| > |X|$  then  $X \leftarrow Y$ 
12:  end if
13: end for
14: return  $X$ 

```

Correctness. Let O be a maximum independent set of G and let $I^* = O \cap S$. Since O is independent, so is I^* in $G[S]$; therefore I^* is one of the sets enumerated in the loop. Consider that iteration.

No vertex of $O \setminus S$ is adjacent to a vertex of I^* , so $O \setminus S$ avoids $N(I^*)$ entirely, hence $O \cap V(G_i)$ is an independent set of $G_i - N(I^*)$ for each i . By induction the recursive call

[figure placeholder]

The recursion picture. A planar graph with the separator S drawn as a red band across the middle and the components $G_1, \dots, G_t$ as blobs on either side. Show one guess $I \subseteq S$ with the chosen vertices circled, their neighbourhood $N(I)$ crossed out in the blobs, and arrows into the blobs labelled “recurse”. A small side note: each blob has at most $9n/10$ vertices and is still planar.

Figure 21: Guess the solution’s trace on the separator; the rest decomposes.

returns a maximum one, so $|I_i| \geq |O \cap V(G_i)|$. Summing,

$$|Y| = |I^*| + \sum_{i=1}^t |I_i| \geq |O \cap S| + \sum_{i=1}^t |O \cap V(G_i)| = |O|.$$

Conversely every Y constructed by the algorithm really is independent: I is independent by the choice of the loop; each I_i is independent; there are no edges between different components G_i, G_j ; and there are no edges between I and any I_i because we deleted $N(I)$ before recursing. So the algorithm outputs an independent set of size at least $|O|$, i.e. a maximum one.

Finally, each subproblem $G_i - N(I)$ is an induced subgraph of a planar graph and hence planar, so the recursion stays inside the class and the separator theorem remains applicable. $\square$

Running time. Let $T(n)$ bound the running time on planar graphs with n vertices. There are at most $2^{|S|} \leq 2^{11\sqrt{n}}$ candidate sets I ; the components number $t \leq n$ and each has at most $\frac{9}{10}n$ vertices, so each recursive call is on an instance of size at most $\frac{9}{10}n$. Absorbing the polynomial work (finding the separator, computing components) into the constants,

$$T(n) \leq 2^{11\sqrt{n}} \cdot n \cdot T\left(\frac{9}{10}n\right) + \text{poly}(n).$$

We claim $T(n) \leq 2^{1000\sqrt{n}}$ for all sufficiently large n . Inductively,

$$T(n) \leq 2^{11\sqrt{n}} \cdot n \cdot 2^{1000\sqrt{9n/10}} = n \cdot 2^{(11+1000\sqrt{9/10})\sqrt{n}} \leq n \cdot 2^{959.7\sqrt{n}} \leq 2^{1000\sqrt{n}},$$

where the last step holds because $n \leq 2^{40\sqrt{n}}$ for every $n \geq 1$. For the base case, brute force costs $2^n \leq 2^{1000\sqrt{n}}$ whenever $n \leq 10^6$, which covers all small instances comfortably. $\square$

Remark 5.2. The constant 1000 is absurd and is an artefact of the crude bookkeeping; the point is only that *some* constant works, giving $2^{\mathcal{O}(\sqrt{n})}$. The same template — separator, guess its trace, recurse — gives $2^{\mathcal{O}(\sqrt{n})}$ algorithms for a whole family of problems on planar graphs, and it is one of the standard reasons to care about Theorem 4.2. Note also that this is not merely a speed-up but a qualitative change: under the Exponential Time Hypothesis, $2^{o(\sqrt{n})}$ is impossible for these problems, so $\sqrt{n}$ in the exponent is the truth.

6 What stays hard on planar graphs

Given Section 5, it is natural to hope that planarity tames NP-hard problems in general. It does not. *Most* NP-hard graph problems remain NP-hard when the input is restricted to planar graphs — including independent set, vertex cover, dominating set, Hamiltonian cycle, and 3-colouring. What we gain is better exponential algorithms, not polynomial ones.

There are a few notable exceptions, and one of them is a pleasant surprise.

6.1 Planar Clique is easy

PLANAR CLIQUE. *Given:* a planar graph G and an integer k . *Asked:* does G contain a clique of size at least k ?

Proposition 6.1. *PLANAR CLIQUE is solvable in polynomial time.*

Proof. A clique of size 5 is a K_5 subgraph, hence a K_5 minor, which is impossible in a planar graph by Corollary 2.5 and Observation 2.8. So every planar graph has clique number at most 4. Therefore: if $k \geq 5$, answer NO immediately; otherwise $k \leq 4$ and we can simply test all $\binom{n}{k} \leq \binom{n}{4} = \mathcal{O}(n^4)$ subsets of size k . $\square$

The moral is worth stating: the problem is not easy because we found a clever algorithm, but because the structure theory forbids the interesting instances from existing at all. When a restriction makes an NP-hard problem polynomial, it is very often for this kind of reason.

6.2 Planar 3-Colouring is NP-hard

Now the other direction, as a sampler of how planar hardness proofs go.

(PLANAR) 3-COLORING. *Given:* a (planar) graph G . *Asked:* does G have a 3-colouring?

We reduce from general 3-COLORING, which is NP-hard. The difficulty is obvious: a general graph need not be planar, and we must somehow simulate the crossings. The tool is a *crossover gadget*, a small planar graph that lets two “wires” pass each other while preserving the colour each carries.

Claim 6.2 (The crossover gadget). *There is a planar graph W with four distinguished corner vertices a, a', b, b' (drawn on the outer face, with a opposite a' and b opposite b') such that:*

- (a) *in any 3-colouring of W , opposite corners get the same colour, i.e. $c(a) = c(a')$ and $c(b) = c(b')$; and*
- (b) *any assignment of colours to the four corners in which opposite corners agree extends to a 3-colouring of all of W .*

The verification is a finite case check, best carried out on the picture: up to permuting the three colours, W admits only two colourings — one in which all four corners get the same colour, and one in which they use two colours (a, a' one colour and b, b' another).

Theorem 6.3. *There is a polynomial-time reduction*

$$3\text{-COLORING} \leq_P \text{PLANAR-3-COLORING}.$$

In particular, PLANAR-3-COLORING is NP-hard.

[figure placeholder]

IMPORTANT — the notes cannot substitute for this picture. Reproduce the gadget W exactly as drawn on the slides: the planar graph with the four corners a, a' (left/right) and b, b' (top/bottom) and its interior triangles. Alongside it, show the two essentially different 3-colourings referred to in Claim 6.2: (i) all four corners the same colour, (ii) a, a' one colour and b, b' a different one. Colour the vertices in the drawing so the case check can be read off directly.

Figure 22: The crossover gadget W and its only two 3-colourings.

Proof. Let G be an instance of 3-COLORING. Draw G in the plane, allowing edges to cross; we may take a drawing in which any two edges cross at most once and no three edges meet at a point, so there are $\mathcal{O}(m^2)$ crossings, which is polynomial.

Now build a planar graph G' : keep all vertices of G , and walk along each drawn edge $\{u, v\}$ from u to v , replacing every crossing it takes part in by a copy of the gadget W , chained so that the colour is handed on from one gadget to the next. Concretely, an edge that participates in ℓ crossings becomes a chain of ℓ gadgets, entering the first at corner a and leaving the last at corner a' , with consecutive gadgets identified at those corners; the final terminal is joined to v by a real edge. The crossing pair of wires enters and leaves the same gadget through b and b' . The result is planar by construction, since each gadget is planar and drawn inside a small disc around the crossing it replaced, and its size is polynomial in $|G|$.

If G is 3-colourable, take a 3-colouring c and give every wire belonging to edge $\{u, v\}$ the colour $c(u)$ along its whole length. At each gadget, the two wires passing through it enter with some colours; opposite corners agree by construction, so by property (b) the colouring extends to the gadget's interior. The final real edge is properly coloured because $c(u) \neq c(v)$. So G' is 3-colourable.

If G' is 3-colourable, restrict the colouring to the wires. By property (a), each gadget forces its opposite corners to agree, so along the chain replacing edge $\{u, v\}$ the colour never changes: the terminal adjacent to v carries the colour $c(u)$. Since that terminal is joined to v by an edge, $c(u) \neq c(v)$. Thus the restriction of the colouring to $V(G)$ is a proper 3-colouring of G .

The construction is clearly computable in polynomial time. $\square$

Intuition. The gadget is doing exactly one job: it is an *insulated crossing*. Two signals need to pass each other in the plane, and the gadget lets them do so while guaranteeing that neither is altered and that they do not interact. Once you have such a component, *any* graph problem whose constraints can be carried along wires becomes as hard on planar inputs as it is in general — which is the real reason so few problems become easy under planarity. Compare this with PLANAR CLIQUE, where the structure theory rules out the hard instances entirely; here the structure theory does not get in the way, so the hardness comes straight through.

[figure placeholder]

Two panels, following the slides. Left: a single crossing of two edges $a-a'$ and $b-b'$, with an arrow to the same picture with the crossing replaced by the gadget W (drawn as a box with four corners labelled). Right: the multiple-crossings case — one long edge from a to a' passing through three boxes labelled W in a row, showing how the colour is relayed along the chain.

Figure 23: Replacing crossings by gadgets, one at a time and in chains.

7 Summary

- **Structure.** Euler's formula $n + f = m + 2$ pins plane drawings down; it gives $m \leq 3n - 6$, the non-planarity of K_5 , and a vertex of degree at most 5 in every planar graph. Planarity is minor-closed and is characterised exactly by forbidding K_5 and $K_{3,3}$ (Kuratowski–Wagner).
- **Colouring.** Six colours by a one-line induction, five by identifying two non-adjacent neighbours of a degree-5 vertex, four by a computer-assisted proof — and three is NP-hard.
- **Testing.** Planarity can be decided in $\mathcal{O}(n^3)$ by a recursion on a separating cycle: segments must go inside or outside, conflicts are recorded in the interlacement graph, and a 2-colouring of that graph makes the choice. Linear-time algorithms exist.
- **Separators.** Every planar graph has a $\frac{9}{10}$ -balanced separator of size $\mathcal{O}(\sqrt{n})$, which is optimal (the grid). Via Koebe's circle packing this follows from a short probabilistic argument.
- **Consequences.** Separators give $2^{\mathcal{O}(\sqrt{n})}$ algorithms, for example for MAXIMUM INDEPENDENT SET. Most NP-hard problems stay NP-hard on planar graphs; CLIQUE is a rare exception, and only because planar graphs have no K_5 .

8 Exercises

Exercise 1. Show that a connected plane graph that is bipartite, simple and has $n \geq 3$ satisfies $m \leq 2n - 4$. Deduce that $K_{3,3}$ is not planar.

Exercise 2. Prove that the number of faces of a plane embedding of a connected planar graph does not depend on the embedding. What goes wrong for disconnected graphs, and what is the right version of Euler's formula for a plane graph with c components?

Exercise 3. Decide whether each of the two graphs in Figure 6 is planar. For the planar one give a crossing-free drawing; for the other exhibit a K_5 or $K_{3,3}$ minor explicitly, listing the branch sets.

Exercise 4. Show that every planar graph has a vertex of degree at most 5, but that degree at most 4 cannot be guaranteed. (Find a planar graph with minimum degree exactly 5.)

Exercise 5. In the proof of Theorem 2.14, where exactly do we use that x and y are non-adjacent? Give an example showing that identifying two *adjacent* neighbours can fail.

Exercise 6. Let C be a cycle and let S, S' be two segments with exactly two attachments each. Show that they conflict in the sense of Definition 3.3 if and only if their attachment pairs interleave around C . Then construct an instance showing that Remark 3.4 is really needed: two segments that do not interleave in the sense of the definition but still cannot be drawn on the same side.

Exercise 7. Show that a cycle plus a single path segment (case (ii) of Lemma 3.7) is always planar, and that it has no separating cycle. Conclude that the exceptional cases in the lemma are genuinely necessary.

Exercise 8. The $\mathcal{O}(n^3)$ bound assumed $m = \mathcal{O}(n)$. Explain precisely why we are entitled to that assumption, and what the algorithm should do on an input with $m > 3n - 6$.

Exercise 9. Adapt Algorithm 3 to compute a maximum *weight* independent set, and to compute a minimum vertex cover. Which parts of the correctness proof change?

Exercise 10. Where does the $2^{\mathcal{O}(\sqrt{n})}$ algorithm break if the separator had size $\mathcal{O}(n^{2/3})$ instead? Work out what running time the same recursion would give.

Exercise 11. Verify Claim 6.2 on the gadget of Figure 22 by hand: enumerate the 3-colourings and check that opposite corners always agree.

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